

CSA 0378-DATA STRUCTURES

20. MERGE SORT

<pre>main.c 1 #include <stdio.h> 2 3 void merge(int a[], int l, int m, int r) { 4 int i=l, j=m+1, k=0, temp[20]; 5 6 while(i<=m && j<=r) { 7 if(a[i] < a[j]) 8 temp[k++] = a[i++]; 9 else 10 temp[k++] = a[j++]; 11 } 12 13 while(i<=m) 14 temp[k++] = a[i++]; 15 16 while(j<=r) 17 temp[k++] = a[j++]; 18 19 for(i=l, k=0; i<=r; i++, k++) 20 a[i] = temp[k]; 21 } 22 23 void mergesort(int a[], int l, int r) { 24 int m; 25 if(l < r) { 26 m = (l+r)/2; 27 mergesort(a, l, m); 28 mergesort(a, m+1, r); 29 merge(a, l, m, r); 30 } 31 } 32 33 int main() { 34 int a[] = {38, 27, 43, 3, 9}; 35 int n = 5, i; 36 37 mergesort(a, 0, n-1); 38 39 printf("Sorted Array: "); 40 for(i=0; i<n; i++) 41 printf("%d ", a[i]); 42 43 return 0; 44 }</pre>	Output Sorted Array: 3 9 27 38 43 *** Code Execution Successful ***
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